

Product Requirements Document

Product: ExamGuard — Smart Exam Allocation & Identity Verification System **Document Version:** 1.0 **Date:** August 14, 2026 **Status:** Draft for Review **Owner:** Product Team

1. Purpose and Background

Educational institutions running large-scale examinations face two persistent operational problems: (1) manually mapping thousands of students to halls, benches, and time slots across multiple exam sessions, and (2) verifying at the door that the person entering a hall is the student who was actually allotted that seat. Paper-based hall tickets are easy to forge, hard to update when rooms change, and provide no audit trail of who actually entered a hall and when.

ExamGuard solves this by digitizing the full exam-day lifecycle: administrators configure exams, halls, and seat maps once; students retrieve a dynamic, time-locked digital pass from any browser; and invigilators verify identity and seat assignment in seconds using a purpose-built mobile scanning app, even without live internet access.

2. Goals and Objectives

- Eliminate manual, spreadsheet-based hall and seat allocation for multi-exam schedules.
- Prevent hall-ticket fraud and impersonation through timed reveal, single-use tokens, and photo-based identity checks.
- Give exam administrators a live, hall-by-hall view of attendance as verification happens.
- Remove the need for students to install a dedicated app — access is via any web browser.
- Allow invigilators to verify students in under 10 seconds per candidate, including in halls with poor or no connectivity.

3. Non-Goals

- ExamGuard does not perform proctoring during the exam itself (no in-exam monitoring, plagiarism detection, or answer-sheet handling).
- ExamGuard does not replace institutional Student Information Systems (SIS); it consumes roster data but is not a system of record for academic history.
- Payment processing, exam-fee collection, and results publishing are out of scope for v1.

4. Target Users and Personas

Persona	Description	Primary Needs
Exam Administrator	College/university exam-cell staff who configure exams, halls, and rosters	Fast bulk setup, error-free seat allocation, real-time visibility
Student	Candidate appearing for one or more exams	Simple access to schedule and pass, no app install, clarity on hall/seat
Invigilator	Faculty or staff assigned to a hall on exam day	Fast, reliable, offline-capable verification with clear visual feedback
Exam Controller / Dean	Senior stakeholder overseeing exam integrity	Aggregate dashboards, audit logs, anomaly/fraud visibility

5. User Stories

Administrator

- As an administrator, I can upload a student roster (CSV/Excel) with roll number, name, email, photo, and department in one operation.
- As an administrator, I can create exam hall records with building, room number, floor, and capacity.
- As an administrator, I can define an exam (course code, title, date, start/end time) and map any number of students to it with hall and seat number, in bulk or individually.
- As an administrator, I can configure the reveal threshold (default T-30 minutes) globally or per exam.
- As an administrator, I can view a live dashboard showing verified, pending, wrong-hall, and absent counts per hall per exam.
- As an administrator, I can export verification and attendance logs for audit and compliance purposes.

Student

- As a student, I can log in with my roll number and password to see all my mapped exams.
- As a student, I can view a dynamic barcode/QR pass for each exam session.
- As a student, I cannot see my hall and seat number until T-30 minutes before the exam start.
- As a student, once the reveal threshold passes, I can see my building, room number, and seat number clearly on my pass.

Invigilator

- As an invigilator, I can scan a student's barcode with my phone camera and instantly see their photo, name, roll number, course, hall, and seat.
- As an invigilator, I get a clear green/red visual and audio cue indicating whether the student is in the correct hall.
- As an invigilator, I can search for a student manually by roll number if their barcode is damaged or their phone is unavailable.
- As an invigilator, I can continue scanning and verifying students even if my phone loses network connectivity, with data syncing automatically once reconnected.
- As an invigilator, I can mark a student absent or flag a discrepancy for administrator review.

6. Functional Requirements

6.1 Admin Web Console

- FR-1: Bulk roster upload via CSV/Excel with validation (duplicate roll numbers, missing photos, malformed emails rejected with a downloadable error report).
- FR-2: CRUD for exam halls (building, room number, floor, capacity).
- FR-3: CRUD for exams (course code, title, date, start time, end time).
- FR-4: Student-exam-hall-seat mapping, supporting both bulk mapping (auto-allocate by capacity) and manual override.
- FR-5: Reveal-threshold configuration, default 30 minutes, adjustable from 5–180 minutes, settable globally or per exam.
- FR-6: Live attendance dashboard with hall-level and exam-level rollups, auto-refreshing via WebSocket.
- FR-7: Exportable verification/audit logs (CSV/PDF) filterable by exam, hall, date range, and status.
- FR-8: Role-based access control for admin users (super admin, exam-cell staff, read-only auditor).

6.2 Student Web Portal

- FR-9: Authentication via roll number and password, with forced password reset on first login.
- FR-10: Timetable view listing all mapped exams with date, time, and course details.
- FR-11: Dynamic barcode/QR generation per exam session, cryptographically signed and time-bound.
- FR-12: Hall and seat details rendered as "Locked" prior to T-30 minutes, auto-unlocking without page reload.
- FR-13: Responsive layout supporting mobile web and desktop browsers.
- FR-14: Barcode is marked "Used" immediately after a successful scan and cannot be reused or shared as a screenshot for entry.

6.3 Invigilator Mobile Scanner App (Expo/React Native)

- FR-15: Camera-based barcode/QR scanning using expo-camera and a barcode-scanning library.
- FR-16: On successful scan, display candidate photo, name, roll number, course, assigned hall, and seat number within 1 second.
- FR-17: Visual feedback: green full-screen flash and confirmation tone for correct hall; red flash with "Wrong Hall" message and correct-hall redirect instructions otherwise.
- FR-18: Manual roll-number search with autocomplete as a fallback verification path.
- FR-19: One-tap "Verify Student" action that logs a VERIFIED status with timestamp and invigilator ID.
- FR-20: Ability to mark ABSENT or FLAGGED with an optional note.
- FR-21: Full offline operation: hall roster and photos pre-cached locally before the exam window; all scans queued locally and synced automatically when connectivity returns.

7. Non-Functional Requirements

Category	Requirement
Performance	Scan-to-result latency under 1 second on-device (offline mode); barcode validation under 300ms server-side (online mode)
Scalability	Support institutions with 50,000+ students and 200+ concurrent exam halls per session
Availability	99.9% uptime for the web portal and API during active exam windows
Security	Signed, time-bound, single-use tokens; encrypted hall/seat data until reveal; TLS 1.2+ everywhere; photos stored in access-controlled object storage
Offline Resilience	Invigilator app must function with zero connectivity for the full duration of a 3-hour exam session
Accessibility	Student web portal meets WCAG 2.1 AA for text contrast and screen-reader compatibility
Auditability	Every verification event is immutably logged with mapping ID, invigilator ID, timestamp, and status
Localization	UI text externalized for future multi-language support

8. Success Metrics

- Reduction in manual seat-allocation effort: target 90% reduction in administrator hours per exam cycle versus spreadsheet-based process.
- Verification speed: median under 10 seconds per student at hall entry, including edge cases.

- Fraud/impersonation incidents: near-zero incidents of wrong-seat or impersonated entry going undetected.
- Adoption: 95%+ of students successfully self-serve their pass without support tickets.
- Reliability: zero data loss on offline-scanned records across a full exam cycle.

9. Assumptions and Constraints

- Institutions can supply digital student photos in a standard format (JPEG/PNG) at roster upload time.
- Exam halls have a one-time setup window before the first exam for the invigilator app to download hall rosters.
- Devices used by invigilators are institution-owned or BYOD smartphones running iOS 13+ or Android 8+, compatible with Expo Go or a built standalone app.
- Internet connectivity, even if intermittent, is available at least once before and after each exam window for sync purposes.

10. Risks

Risk	Mitigation
Students screenshot/share barcodes	Time-bound signed tokens, single-use marking, optional live liveness check in future release
Bulk roster data quality issues	Strict validation with a clear pre-upload error report and dry-run mode
Invigilator device failure on exam day	Manual roll-number search fallback; spare pre-provisioned devices per hall
Reveal-time clock drift between server and client	Server-authoritative reveal timing enforced via signed, server-issued unlock events, not client clocks

11. Release Scope (v1)

In scope: admin console, student web portal, invigilator Expo app, single-institution deployment, CSV roster import, offline scanning, live dashboard.

Out of scope for v1: multi-institution SaaS tenancy, biometric/face-match automation, payment integration, native (non-Expo) app store builds, SIS two-way sync.